Further Practice: Unconfoundedness

BUSS975 — covering Lectures 3a and 3b

*These problems are **optional**. They are not collected and not graded. They extend the assigned set (`03_unconfoundedness_PS.qmd`) and are good preparation for the comprehensive exam. Each stands on its own.*

Problem E1 (Covariate selection without a DAG)

For each candidate control, classify it as a **confounder**, a **mediator**, a **collider**, or **neutral**, and state whether including it helps, hurts, or is merely optional. Where a variable could be more than one thing, say what would settle it.

(a) D = index inclusion, Y = investment, candidate = institutional ownership measured *after* inclusion.

(b) D = bank recapitalization, Y = lending, candidate = the bank's pre-crisis capital ratio.

(c) D = adoption of a governance provision, Y = firm value, candidate = whether the firm was later acquired.

(d) D = a randomized disclosure mandate, Y = cost of capital, candidate = analyst coverage, which predicts Y strongly and is unrelated to D.

(e) Cunningham's rule of thumb is to control for variables determined **before** treatment. Give a case where that rule keeps a bad control anyway, and a case where it discards a good one.

Problem E2 (The two assumptions differ in kind)

(a) State unconfoundedness and common support. Explain why exactly one of them is testable, and what specifically you can and cannot verify about the other.

(b) The lecture claims the ATE requires the *strong* form of unconfoundedness and the ATT only a weaker one. Write both versions formally and show which term in each identification argument uses which.

(c) Under **weak** unconfoundedness, is it still true that $ATE(x) = ATT(x)$? Prove or give a counterexample.

(d) A paper reports: "We trim observations with $\hat{e} < 0.05$ or $\hat{e} > 0.95$, leaving 87% of the sample." State precisely what the reported estimate is now an estimate *of*, and draft the one sentence the paper should add to its results section.

(e) Imbens gives three defences of unconfoundedness-based methods. Reconstruct them, and say which you find least persuasive and why.

Problem E3 (Subclassification and the curse of dimensionality)

(a) Using R's built-in `Titanic` table, reproduce the raw first-class survival gap, then the subclassification estimate stratifying on sex and age. Report the ATE, ATT and ATU and explain why they differ.

(b) Explain why the ATU is the least trustworthy of the three in this application.

(c) With k binary covariates the number of strata is 2^k. At $n = 5{,}000$, how many covariates can you stratify on before the average group holds fewer than five treated units? Simulate the collapse: report the share of strata with no treated or no control unit as k rises.

(d) Cochran's method and exact matching are the same idea executed differently. State the one operational difference, and say which of the two makes the common-support failure more visible.

Problem E4 (Bias correction, worked)

Ten workers were eligible for a training program. Four enrolled.

Worker	Andy	Betty	Chad	Doug
Age	23	27	29	22
Earnings (Y^1)	\$200	\$250	\$150	\$300

Worker	Edith	Fred	Gina	Hank	Inez	Janet
Age	27	32	26	26	28	29
Earnings (Y^0)	\$225	\$500	\$200	\$190	\$180	\$140

(a) Match on age, one neighbour, averaging ties. Report each $\hat{\delta}_i$ and $\widehat{ATT}$.

(b) Regress earnings on age using the **matched controls only**, and report $\hat{\alpha}$ and $\hat{\beta}$. Then compute the bias-corrected effects and $\widehat{ATT}^{BC}$.

(c) Refit step (b) on **all six** controls and recompute. The answer changes enormously. Identify the single observation responsible and explain what it demonstrates about the method's exposure.

(d) Why do the exactly-matched workers receive a correction of zero? Show it algebraically for the one-covariate linear case.

(e) Abadie–Imbens's theory uses all controls in step 2; every implementation uses matched controls only. Give an argument for each choice.

Problem E5 (Regression, matching, and what neither can do)

(a) State the Frisch–Waugh–Lovell theorem in its three steps. Cunningham's `fwl.do` residualizes the **regressor only**; the standard statement residualizes **both sides**. Show that the two return the same $\hat{\beta}_1$, but that only the double-residual version reproduces the full regression's residuals. Then explain why the single-residual standard error is simply wrong, while the double-residual one is too small by exactly $\sqrt{(n-2)/(n-p)}$ — and say where that factor comes from.

(b) The section opens by calling matching an **interpolation** method and regression an **extrapolation** one. Say what each does when a treated unit has no comparable control. Then state what regression buys with that extrapolation and what it pays for it, and explain why "regression needs no common support" is a claim about the estimator rather than about the data.

(c) Regression adjustment (outcome imputation) and IPW use two different models. State what each assumes, then explain what a doubly robust estimator buys and what it does *not* buy.

(d) "Matching creates no new variation." Explain what this means operationally, and use it to answer: can matching fix an unobserved confounder, reverse causality, or a bad control? Answer each separately.

(e) LaLonde (1986), Dehejia and Wahba (1999), and Smith and Todd (2005) reached three different verdicts on the same data. State each verdict in one sentence, and say what the sequence as a whole establishes about propensity score methods.

Problem E6 (Horvitz–Thompson versus Hájek)

(a) Write both estimators of $E[Y^1]$. Show that replacing Y_i by $Y_i + c$ changes the HT estimate by $c\,\hat{1}_T$, where $\hat{1}_T = \frac{1}{N}\sum_i D_i/\hat{e}_i$, but changes the Hájek estimate by exactly c. Name the property this establishes, and say why it is the reason to prefer Hájek (Ding 2024, Proposition 11.1).

(b) Four treated banks have $\hat{e} = (0.5, 0.4, 0.25, 0.2)$ and **every one has** $Y = 20$. With $N = 10$, compute both estimates. No score is anywhere near 0 or 1, yet HT is 35% too high. Identify the quantity that fails to equal what HT assumes about it.

(c) Now suppose the fourth bank's score was 0.02 and its outcome 44, the others unchanged. Recompute both estimates and report the largest single weight share. One of the two estimates lies **outside the range of the observed data** — say which, and explain why that is a worse failure than merely being biased.

(d) In (c) the Hájek estimate is defensible arithmetic but not a reliable estimate. Explain the distinction, and name the diagnostic you would report beside it. Then explain why retreating to the ATT removes this particular problem, and construct a case in which it would not.